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Numerical and experimental study of the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a poro-mechanical approach

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Abstract

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This thesis investigates the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a poro-mechanical approach.

The study combines numerical modeling and experimental validation to better understand the coupled physics involved.

Nomenclature

Finite Element notation

$[\mathbf{B}]$	Compatibility (strain–displacement) matrix	
$[\mathbf{C}]$	Capacity matrix	
$[\mathbf{K}]$	Global stiffness matrix	
$[\mathbf{K}_T]$	Thermal conductivity matrix	
$[\mathbf{N}]$	Shape function matrix	
Δt	Time step	[s]
θ	Wilson time integration parameter	[-]
$\{\mathbf{F}\}$	Nodal force / flux vector	
$\{\mathbf{R}\}$	Internal force (residual) vector	
$\{\mathbf{U}\}$	Nodal displacement vector	
$\{T\}$	Nodal temperature vector	

Thermal symbols

β_s	Thermal dilation coefficient of the solid skeleton	[K^{-1}]
ΔH_{vap}	Enthalpy of evaporation	[$\mathrm{J{\cdot}kg}^{-1}$]
λ	Thermal conductivity of the porous medium	[$\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$]
λ_d	Thermal conductivity of the dry skeleton	[$\mathrm{W{\cdot}m^{-1}{\cdot}K^{-1}}$]

λ_{d0}	Reference dry thermal conductivity	[W·m^{-1}·K^{-1}]
ρC_p	Overall volumetric heat capacity of the porous medium	[J·m^{-3}·K^{-1}]
ρ	Mass density	[kg·m^{-3}]
ρ_a	Dry air density	[kg·m^{-3}]
ρ_g	Gas mixture density	[kg·m^{-3}]
ρ_l	Liquid water density	[kg·m^{-3}]
ρ_s	Solid skeleton density	[kg·m^{-3}]
ρ_v	Water vapor density	[kg·m^{-3}]
σ_{SB}	Stefan–Boltzmann constant	[W·m^{-2}·K^{-4}]
ε_{em}	Emissivity	[-]
A_λ	Thermal conductivity temperature coefficient	[K^{-1}]
C_g	Specific heat capacity of gas mixture (advection term)	[J·kg^{-1}·K^{-1}]
C_l	Specific heat capacity of liquid (advection term)	[J·kg^{-1}·K^{-1}]
c_p	Specific heat capacity	[J·kg^{-1}·K^{-1}]
C_{pa}	Specific heat capacity of dry air	[J·kg^{-1}·K^{-1}]
C_{pl}	Specific heat capacity of liquid water	[J·kg^{-1}·K^{-1}]
C_{ps}	Specific heat capacity of the solid skeleton	[J·kg^{-1}·K^{-1}]
C_{pv}	Specific heat capacity of water vapor	[J·kg^{-1}·K^{-1}]
h	Convective heat transfer coefficient	[W·m^{-2}·K^{-1}]
H_{dehyd}	Enthalpy of dehydration	[J·kg^{-1}]
q''	Heat flux	[W·m^{-2}]
T	Temperature	[K]
T_0	Reference temperature	[K]
T_{crit}	Critical temperature of water (647 K)	[K]

Hydraulic symbols

Δm_{dehyd}	Cumulative dehydrated water mass	[kg·m^{-3}]
$\dot{m}_{dehyd}$	Dehydration rate (negative, mass loss of solid)	[kg·m^{-3}·s^{-1}]
$\dot{m}_{vap}$	Evaporation rate (positive for evaporation)	[kg·m^{-3}·s^{-1}]
$\mathbf{v}_{g-s}$	Velocity of gas relative to the solid skeleton	[m·s^{-1}]
$\mathbf{v}_{l-s}$	Velocity of liquid relative to the solid skeleton	[m·s^{-1}]
μ_g	Dynamic viscosity of gas mixture	[Pa·s]
μ_l	Dynamic viscosity of liquid water	[Pa·s]
ϕ	Porosity	[-]
ϕ_0	Initial porosity	[-]
τ	Tortuosity factor	[-]
a	Van Genuchten pressure parameter	[Pa]
A_ϕ	Porosity temperature coefficient	[K^{-1}]
A_D	Permeability–damage coefficient	[-]
A_p	Permeability pressure exponent	[-]
A_S	Van Genuchten exponent for relative permeability	[-]
A_T	Permeability temperature coefficient	[K^{-1}]
b	Van Genuchten shape parameter	[-]
D	Effective diffusivity of vapor in the porous medium	[m^2·s^{-1}]
D_{v0}	Reference binary diffusion coefficient	[m^2·s^{-1}]
D_{va}	Binary diffusion coefficient of vapor in air	[m^2·s^{-1}]
K	Intrinsic permeability	[m^2]
K_0	Reference intrinsic permeability	[m^2]
k_{rg}	Relative permeability of gas	[-]
k_{rl}	Relative permeability of liquid	[-]
M_a	Molar mass of dry air	[kg·mol^{-1}]
m_a	Mass of dry air per unit volume	[kg·m^{-3}]
M_g	Molar mass of gas mixture	[kg·mol^{-1}]

m_l	Mass of liquid water per unit volume	[kg·m^{-3}]
m_s	Mass of solid per unit volume	[kg·m^{-3}]
M_v	Molar mass of water vapor	[kg·mol^{-1}]
m_v	Mass of water vapor per unit volume	[kg·m^{-3}]
p_a	Partial pressure of dry air	[Pa]
p_c	Capillary pressure	[Pa]
p_g	Gas pressure	[Pa]
p_l	Liquid water pressure	[Pa]
p_s	Average pore pressure acting on the solid skeleton	[Pa]
p_v	Partial pressure of water vapor	[Pa]
p_{atm}	Atmospheric pressure	[Pa]
p_{vs}	Saturation vapor pressure	[Pa]
R	Universal gas constant	[J·mol^{-1}·K^{-1}]
S_l	Degree of liquid saturation	[-]

Mechanical symbols

α	Biot's coefficient	[-]
α_c	Mazars compression weight	[-]
α_t	Mazars tension weight	[-]
$\boldsymbol{\sigma}$	Cauchy stress tensor (undamaged)	[Pa]
$\boldsymbol{\sigma}'$	Effective stress tensor (skeleton)	[Pa]
$\boldsymbol{\sigma}^{Total}$	Total stress tensor	[Pa]
$\boldsymbol{\varepsilon}$	Total strain tensor	[-]
$\boldsymbol{\varepsilon}_{\sigma}$	Mechanical strain (elastic + creep)	[-]
$\boldsymbol{\varepsilon}_e$	Elastic strain tensor	[-]
$\boldsymbol{\varepsilon}_{sh}$	Capillary shrinkage strain	[-]

$\boldsymbol{\varepsilon}_{th}$	Thermal dilation strain	[-]
$\mathbb{C}$	Fourth-order isotropic elasticity tensor	[Pa]
$\mathcal{D}$	Isotropic damage variable	[-]
ν	Poisson's ratio	[-]
ε_{d0}	Initial damage threshold strain	[-]
ε_{eq}	Mazars equivalent strain	[-]
A_c	Mazars softening parameter (compression)	[-]
A_t	Mazars softening parameter (tension)	[-]
B_c	Mazars softening exponent (compression)	[-]
B_t	Mazars softening exponent (tension)	[-]
B_t^{mod}	Regularized Mazars tension exponent	[-]
E	Young's modulus	[Pa]
f_t	Tensile strength	[Pa]
G_f	Fracture energy	[J·m^{-2}]
L_e	Characteristic element length	[m]

Probabilistic symbols

β	Generalised reliability index $= -\Phi^{-1}(P_f)$
δ_3	Skewness $= \mu_3/\sigma^3$ (third standardised moment)
δ_4	Kurtosis $= \mu_4/\sigma^4$ (fourth standardised moment)
$\hat{P}_f$	Estimated probability of failure (Monte Carlo)
$\mathbf{X}$	Vector of random variables $(X_1, \ldots, X_N)$
μ	Mean (first moment)
Φ	Standard normal CDF
Φ^{-1}	Inverse standard normal CDF
ρ_H	Autocorrelation function of field H

σ	Standard deviation	
σ^2	Variance (second central moment)	
CoV	Coefficient of variation $= \sigma/\mu$	
C_H	Covariance function of field H	
C_i	Contribution of X_i to output variance (%)	
D_f	Failure domain ($G \leq 0$)	
D_s	Safety domain ($G > 0$)	
e	Acceptable absolute error (Monte Carlo convergence)	
F_X	Cumulative distribution function (CDF) of X	
f_X	Probability density function (PDF) of X	
G	Model output; performance function in reliability ($G > 0$: safe)	
$g(\cdot)$	Deterministic model (input–output mapping)	
$H(\mathbf{x})$	Random field	
K	Number of quadrature points per variable	
L_c	Correlation length	[m]
L_{struct}	Characteristic structural dimension	[m]
n	Number of Monte Carlo simulations	
N_f	Number of failure realisations	
P_f	Probability of failure	
S_i	First-order Sobol index of X_i	
S_{ij}	Second-order Sobol index (interaction between X_i and X_j)	
S_{T_i}	Total-order Sobol index of X_i	
V_i	First-order partial variance of X_i	
V_{ij}	Second-order partial variance (interaction X_i, X_j)	
$W(x)$	Weight function ($=$ PDF f_X) in quadrature integral	
w_k	Quadrature weight	
X	Random variable	
x_k	Quadrature abscissa (evaluation point)	

Contents

Chapter 1

Introduction

[REVIEW] This chapter is currently a draft version and will undergo further review and revision.

1.1 Context and problem statement

Concrete structures exposed to fire undergo a complex interplay of thermal, hydric, and mechanical phenomena that can lead to progressive degradation and, in the most severe cases, to *spalling* — the sudden detachment of concrete layers from the exposed surface. Several catastrophic tunnel fires have illustrated the severity of this phenomenon: the Channel Tunnel fire (1996), the Mont-Blanc tunnel fire (1999), and the St. Gotthard tunnel fire (2001) all caused extensive spalling-induced structural damage [5].

In nuclear containment structures, where concrete is exposed to extreme temperatures during severe accidents, the consequences of spalling are particularly critical as they may compromise structural integrity and containment function.

1.1.1 Concrete behaviour under high temperature

When concrete is subjected to rapid heating, several coupled physical processes develop simultaneously:

- **Thermal:** steep temperature gradients develop through the cross-section, driving differential thermal dilation and generating compressive stresses at the heated surface.
- **Hydric:** free and chemically bound water evaporate and migrate through the pore network; in low-permeability concretes such as high-performance concrete (HPC),

vapor cannot escape rapidly enough, leading to pore pressure build-up behind a quasi-saturated *moisture clog.*

- **Mechanical:** the combined action of thermal stresses and pore pressure loading on the solid skeleton can exceed the (temperature-degraded) tensile strength, initiating cracks parallel to the heated surface and potentially triggering explosive spalling.

Despite extensive experimental research, the mechanisms governing spalling remain incompletely understood and no general consensus has been reached [5].

1.1.2 Problem statement

Reliable prediction of spalling requires a consistent *thermo-hydro-mechanical* (THM) framework capable of capturing:

- the multi-physics coupling between thermal, hydric, and mechanical fields under high temperature;
- the temperature- and damage-dependent evolution of transport properties, whereby microcracking alters permeability and redistributes pore pressure;
- the inherent spatial variability of material parameters (permeability, porosity, tensile strength) and its influence on the onset and location of spalling.

1.1.3 Limitations of existing THM models

Current THM models present several limitations that this work aims to address:

- incomplete coupling between the damage state and transport properties, often neglecting the feedback of cracking on permeability;
- lack of consistent regularization for strain localization, leading to mesh-dependent solutions;
- limited probabilistic assessment of parameter uncertainty: existing models typically treat material properties as homogeneous deterministic values, neglecting the spatial variability of concrete properties.

1.2 Objectives and scope of the thesis

Main objectives of the thesis:

- Understanding and Developing a THM model
- Temperature- and damage-dependent evolution of stiffness and transport properties.
- Numerical implementation and numerical validation.
- Probabilistic sensitivity analysis: Evaluating parameter uncertainty and identifying the dominant physical parameters.

The scope of this work is limited to:

- Thermal boundaries limited by severe accident conditions.
- Continuum-scale THM modelling by Cast3M.
- Damage modelling including temperature-dependent stiffness and transport properties.
- Probabilistic sensitivity analysis at the material parameter level.

1.3 Research enviroment and Computational tools

1.3.1 3SR Laboratory

The host institution providing the scientific environment, experimental data, and expertise in structural mechanics and risk analysis.

1.3.2 Finite element code - Cast3M

Cast3M (or Castem) is an open-source finite-element code developed at CEA, which is used to implement advanced constitutive laws and custom multiphysics formulations.

Compared with COMSOL Multiphysics, Cast3m typically offers greater transparency and control for implementing tightly coupled THM while COMSOL offers a user-friendly GUI for many built-in multiphysics interfaces.

1.3.3 Probabilistic analysis tool - Persalys

Persalys is the computational tool dedicated to managing parameter uncertainty in material properties.

It is used to perform the probabilistic sensitivity analysis, generate random sampling (e.g., Monte Carlo simulations), and rank the dominant THM parameters.

Chapter 2

Theoretical and Numerical framework

[REVIEW] This chapter is currently a draft version and will undergo further review and revision.

2.1 Thermal behavior

2.1.1 Conduction, convection and radiation

Heat transfer is the transport of thermal energy driven by a temperature difference. The following definitions of heat transfer modes are summarized from standard references [3].

Conduction

Conduction is the transfer of energy from the more energetic to the less energetic particles of a substance

- **Physical Mechanism:**
- **Rate Equation (Fourier's Law):** For a one-dimensional plane wall having a temperature distribution $T(x)$, the rate equation is expressed as Fourier's law:

$$q''_{cond} = -\lambda \frac{dT}{dx} \tag{2.1}$$

Where q''_{cond} is the heat flux (W/m^2), λ is the thermal conductivity $(W/m \cdot K)$, and the minus sign indicates that heat is transferred in the direction of decreasing temperature.

Convection

The convection heat transfer mode is

- **Physical Mechanism:** Convection heat transfer occurs between a fluid in motion and a bounding surface when the two are at different temperatures
- **Rate Equation (Newton's Law of Cooling):** Regardless of the nature of the convection process, the appropriate rate equation is:

$$q''_{conv} = h(T_s - T_\infty) \tag{2.2}$$

Where q'' is the convective heat flux (W/m^2), h is the convection heat transfer coefficient $(W/m^2 \cdot K)$, T_s is the surface temperature, and T_∞ is the fluid temperature.

Radiation

Thermal radiation is energy emitted by matter that is at a nonzero temperature, and unlike conduction and convection, it does not require the presence of a material medium.

- **Physical Mechanism:** The energy of the radiation field is transported by electromagnetic waves (or photons)
- **Emissive Power (Stefan-Boltzmann law):** The upper limit to the emissive power (rate of energy released per unit area) for an ideal radiator or blackbody is given:

$$E_b = \sigma T_s^4 \tag{2.3}$$

The heat flux emitted by a real surface is given by $E = \varepsilon\sigma T_s^4$, where σ is the Stefan-Boltzmann constant and ε is the emissivity.

- **Net Radiation Exchange:** For a gray surface at T_s completely surrounded by a much larger isothermal surface at T_{sur}, the net rate of radiation heat transfer per unit area is:

$$q''_{rad} = \varepsilon\sigma(T_s^4 - T_{sur}^4) \tag{2.4}$$

2.1.2 Heat transfer

Heat Equation

Based on the principle of energy conservation and Fourier's law, the general heat diffusion equation (or heat equation) for a medium is detailed in [7] and can be expressed as:

$$\rho c_p \frac{\partial T}{\partial t} + \text{div}(-\lambda \overrightarrow{\text{grad}}(T)) - q = 0 \tag{2.5}$$

Where:

- ρ is the volumetric mass density (kg/m^3).
- c_p is the specific heat capacity ($J/kg \cdot K$).
- λ is the thermal conductivity ($W/m \cdot K$).
- T is the temperature field and t is time.
- q is the volume heat flux source (W/m^3).

To solve this equation, appropriate boundary conditions must be specified:

- **Prescribed temperatures :** The temperature is specified as $T = T_{imp}$ on the boundary surface ∂V^T.
- **Prescribed surface heat flux :** The total heat flux due to conduction, convection, and radiation on the boundary surface ∂V^φ is given by:

$$\vec{n} \cdot (\lambda \overrightarrow{\text{grad}}(T)) = \varphi_{imp} + h(T_f - T) + \varepsilon\sigma(T_\infty^4 - T^4) \tag{2.6}$$

where φ_{imp} is the prescribed heat flux, $h(T_f - T)$ represents the convection term, and $\varepsilon\sigma(T_\infty^4 - T^4)$ is the radiation term.

Thermal properties

The thermal properties of concrete are strongly influenced by both the hydric state and the material composition.

The overall heat capacity ρC_p reflects the multiphase nature of the porous medium, combining contributions from the solid skeleton, liquid water, dry air, and water vapor [**lewis1998**]:

$$\rho C_p = (1-\phi)\,\rho_s\,C_{ps} + \phi\,[S_l\,\rho_l\,C_{pl} + (1-S_l)\,(\rho_a\,C_{pa} + \rho_v\,C_{pv})] \tag{2.7}$$

where:

The solid density decreases linearly with temperature due to dehydration: $\rho_s(T) = \rho_{s0}[1 - A_\rho(T - T_0)]$ with $\rho_{s0} = 2611$ kg/m^3.

The liquid water density evolves with temperature according to a polynomial fit [**furbish1997**]:

$$\rho_l(T) = \sum_{i=0}^{5} a_i \, T^i \tag{2.8}$$

with coefficients a_i given in Table **??**.

The solid thermal capacity evolves linearly with temperature [**harmathy1973**]:

$$C_{ps}(T) = C_{ps0}[1 + A_c(T - T_0)] \tag{2.9}$$

with $C_{ps0} = 940$ J·kg^{-1}K^{-1}.

The thermal conductivity depends on saturation, porosity, and temperature [**baggio1993**]:

$$\lambda = \lambda_d \left(1 + 4 \, \frac{S_l \, \phi \, \rho_l}{(1 - \phi) \, \rho_s} \right) \tag{2.10}$$

with [**harmathy1973**]:

$$\lambda_d = \lambda_{d0} \left[1 + A_\lambda \, (T - T_0) \right] \tag{2.11}$$

As drying progresses ($S_l \to 0$), both ρC_p and λ decrease, which accelerates the penetration of the thermal front into the material. These dependencies highlight the influence of the hydric state on the thermal field, which is further discussed in Section 2.4.1.

Discretization

Using the FE discretization, the temperature and temperature gradient in the spatial domain are expressed in matrix form as:

$$T(x) = [N(x)]\{T\}, \quad \overrightarrow{\text{grad}}(T) = [B(x)]\{T\} \tag{2.12}$$

From the weak formulation of the heat equation combined with the spatial discretization, we obtain the general matrix equation:

$$[C]\{\dot{T}\} + [K]\{T\} = \{F\} \tag{2.13}$$

Where the matrices are defined as follows:

- **Capacity matrix:**

$$[C] = \int_V \rho c_p [N]^T [N] dV \quad (J/K) \tag{2.14}$$

- **Conductivity matrix:**

$$[K] = \int_V [B]^T[\lambda][B]dV + \int_{\partial V\varphi} h[N]^T[N]dS \quad (W/K) \tag{2.15}$$

- **Equivalent nodal flux vectors:**

$$\{F\} = \int_V [N]^T q dV + \int_{\partial V\varphi} [N]^T \left(\varphi_{imp} + hT_f + \varepsilon\sigma(T_\infty^4 - T^4)\right) dS \quad (W) \tag{2.16}$$

For steady-state thermal analysis, the temperature is time-independent ($\{\dot{T}\} = 0$) (assuming constant material properties and neglecting radiation):

$$[K]\{T\} = \{F\} \tag{2.17}$$

By solving this linear system, we can determine the unknown temperatures $\{T\}$ at the nodes across the entire global mesh.

2.1.3 Thermal gradient

Due to the low thermal conductivity of concrete, fire exposure produces steep temperature gradients near the heated surface. The hot surface layer tends to expand while the cooler interior restrains this deformation, generating compressive stresses at the surface and tensile stresses in the interior.

Since concrete is inherently weak in tension, this stress state can initiate cracking parallel to the heated face. The role of thermal gradient in the spalling process is discussed further in Section 2.5.2.

2.2 Hydric behavior

[REVIEW]

When concrete is exposed to high temperatures, the behavior of water within the porous network plays a central role in the thermo-hydro-mechanical response of the material. This section presents the classification of water in concrete, the phase-change mechanisms, the governing equation for moisture transport, and the constitutive laws required to close the system.

2.2.1 Types of water in concrete

Concrete is a multiphase porous material whose voids can be filled with water, air and other fluids. The water present in the cement paste can be broadly classified into two categories [5]

- **Pore water (free/evaporable water):** This water evaporates when the thermodynamic conditions are met (*i.e.*, at $T = 100\,°\mathrm{C}$ and atmospheric pressure) and exists in three main forms:
  - *Capillary and inter-hydrate water:* fills the capillary pores (size $\sim$ 100s of nm) and inter-hydrate pores(size $\sim$ 10s of nm), corresponding to a saturation $S_{ssp} < S_l < 1$.
  - *Physically adsorbed water (gel water):* retained by the surface of the solid matrix due to Van der Waals forces, confined in gel pores (size $\sim$ a few nm).
  - *Interlayer water:* confined in $\sim$ 1 nm space between the C-S-H layers.
- **Chemically bound water (non-evaporable):** This water is combined with anhydrous cement during hydration and forms part of the crystalline structure of hydration products (e.g., C-S-H, portlandite). It can only be released through thermal decomposition (dehydration) at elevated temperatures.

It is important to note that the range of pore sizes is a continuous one; the dimensional boundaries between capillary, inter-hydrate and gel pores are approximations to inherently blurred boundaries.

2.2.2 Evaporation and dehydration

Two distinct phase-change mechanisms govern the release of water in heated concrete:

- **Evaporation** is the transition of free liquid water into vapor. It occurs progressively as temperature increases, becoming significant above approximately 100 °C at atmospheric pressure. The evaporation rate $\dot{m}_{vap}$ is determined from the mass conservation of liquid water (see Section 2.2.3).
- **Dehydration** is the thermally-driven decomposition of hydration products, releasing chemically bound water as vapor. The main reactions occur at characteristic temperature ranges [5]:
  - Up to 300 °C: progressive decomposition of C-S-H phases and ettringite;
  - 400–500 °C: dehydroxylation of portlandite ($Ca(OH)_2 \rightarrow CaO + H_2O$);

– $\sim 570\,^\circ$C: α–β quartz transformation;
– Above $600\,^\circ$C: decarbonation of calcareous aggregates ($CaCO_3 \rightarrow CaO + CO_2$).

Both evaporation and dehydration act as vapor sources that increase gas pressure within the pore network. They also consume latent energy, affecting the temperature field (see Section 2.4).

Dehydration kinetics

The cumulative dehydrated water mass is described by the following phenomenological relationship [**pesavento2000**]:

$$\Delta m_{dehyd} = f_s \cdot f_m \cdot f_c \cdot f(T) \tag{2.18}$$

where f_s is a stoichiometric factor, f_m accounts for the concrete's age, f_c is the cement mass per unit volume, and $f(T)$ is a temperature-dependent function:

$$f(T) = \begin{cases} 0 & \text{if } T < 105\,^\circ\text{C} \\ \left[1 + \sin\left(\frac{\pi}{2}\,(1 - 2\exp(-0.004\,(T - 105)))\right)\right] & \text{if } T \geq 105\,^\circ\text{C} \end{cases} \tag{2.19}$$

Enthalpy of evaporation

The enthalpy of evaporation is temperature-dependent and is approximated by the Watson formula [**reid1987**]:

$$\Delta H_{vap}(T) = \Delta H_{vap,0} \left(\frac{T_{crit} - T}{T_{crit} - T_0}\right)^{0.38} \tag{2.20}$$

where $T_{crit} = 647$ K is the critical temperature of water and $\Delta H_{vap,0} = 2.672 \times 10^5$ J/kg.

2.2.3 Moisture transport equation

Transport mechanisms

The transport of moisture in heated concrete is governed by two mechanisms:

- **Advective flow (Darcy's law):** The bulk flow of liquid water and gas through the porous network is driven by pressure gradients:

$$\mathbf{v}_{l-s} = -\frac{K\,k_{rl}}{\mu_l}\,(\nabla p_g - \nabla p_c) \tag{2.21}$$

$$\mathbf{v}_{g-s} = -\frac{K\,k_{rg}}{\mu_g}\nabla p_g \tag{2.22}$$

where K is the intrinsic permeability, k_{rl} and k_{rg} are the relative permeabilities of liquid and gas, μ_l and μ_g are the dynamic viscosities, and $p_l = p_g - p_c$ is the liquid pressure.

- **Diffusive flow (Fick's law):** Water vapor diffuses within the gas mixture according to:

$$\mathbf{j}_v = -\rho_g\, D\, \nabla\left(\frac{\rho_v}{\rho_g}\right) \tag{2.23}$$

where D is the effective diffusivity accounting for the tortuous porous network.

Mass conservation equations

The mathematical framework for moisture transport in heated concrete is based on the conservation of mass for each phase. The general form of a mass conservation equation for a phase π is [5]:

$$\frac{\partial}{\partial t}(\text{density}) + \nabla\cdot(\text{flux}) = \text{source} \tag{2.24}$$

Concrete at high temperature is treated as a multiphase porous medium consisting of a solid skeleton (s), liquid water (l), water vapor (v), and dry air (a). Applying Equation (2.24) to each phase yields [5]:

- **Solid skeleton:**

$$\frac{\partial m_s}{\partial t} = \dot{m}_{dehyd} \tag{2.25}$$

- **Liquid water:**

$$\frac{\partial m_l}{\partial t} + \nabla\cdot(m_l\,\mathbf{v}_{l-s}) = -\dot{m}_{vap} - \dot{m}_{dehyd} \tag{2.26}$$

- **Water vapor:**

$$\frac{\partial m_v}{\partial t} + \nabla\cdot(m_v\,\mathbf{v}_{g-s}) + \nabla\cdot(m_v\,\mathbf{v}_{v-g}) = \dot{m}_{vap} \tag{2.27}$$

- **Dry air:**

$$\frac{\partial m_a}{\partial t} + \nabla\cdot(m_a\,\mathbf{v}_{g-s}) + \nabla\cdot(m_a\,\mathbf{v}_{a-g}) = 0 \tag{2.28}$$

where $\mathbf{v}_{\pi-s}$ is the velocity of phase π ($\pi = l, g$) relative to the solid skeleton, and $\mathbf{v}_{\pi-g}$ is the velocity of species π ($\pi = v, a$) relative to the gas mixture. The source term $\dot{m}_{vap}$ represents the evaporation rate ($\dot{m}_{vap} > 0$ for evaporation, < 0 for condensation), and $\dot{m}_{dehyd}$ is the dehydration rate ($\dot{m}_{dehyd} < 0$, treated as a mass loss of the solid skeleton and a mass source for liquid water).

The mass of each phase per unit volume of porous medium is [5]:

$$m_s = (1-\phi)\,\rho_s \tag{2.29}$$

$$m_l = \rho_l\, S_l\, \phi \tag{2.30}$$

$$m_v = \rho_v\,(1-S_l)\,\phi \tag{2.31}$$

$$m_a = \rho_a\,(1-S_l)\,\phi \tag{2.32}$$

where ϕ is the porosity and S_l is the liquid saturation degree.

By combining the mass conservation of liquid water and water vapor, the evaporation source term $\dot{m}_{vap}$ is eliminated, yielding the **water species conservation equation**:

$$\begin{aligned} & S_l\rho_l\frac{\partial \phi}{\partial t} + S_l\phi\frac{\partial \rho_l}{\partial t} + \rho_l\phi\frac{\partial S_l}{\partial t} + (1-S_l)\phi\frac{\partial \rho_v}{\partial t} - \rho_v\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_v\frac{\partial \phi}{\partial t} \\ & + \nabla\cdot\left(-K\frac{\rho_l k_{rl}}{\mu_l}\nabla p_l\right) + \nabla\cdot\left(-K\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_v}{p_g}\right) = -\dot{m}_{dehyd} \end{aligned} \tag{2.33}$$

This equation has three primary unknowns: the temperature T, the capillary pressure p_c, and the gas pressure p_g. Similarly, the **dry air conservation equation** reads:

$$(1-S_l)\phi\frac{\partial \rho_a}{\partial t} - \rho_a\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_a\frac{\partial \phi}{\partial t} + \nabla\cdot\left(-K\frac{\rho_a k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_a}{p_g}\right) = 0 \tag{2.34}$$

Constitutive relations for the gas phase

The gaseous phase (dry air + water vapor) is governed by the following relations [5]:

- **Ideal gas law:**

$$\rho_\pi = \frac{p_\pi M_\pi}{RT}, \quad \pi = a, v \tag{2.35}$$

  where M_π is the molar mass and R is the universal gas constant.

- **Dalton's law:**

$$p_g = p_a + p_v \tag{2.36}$$

- **Kelvin equation** (thermodynamic equilibrium between liquid and vapor):

$$p_v = p_{vs}(T)\,\exp\left(\frac{M_v}{\rho_l RT}(p_g - p_c - p_{vs})\right) \tag{2.37}$$

  where $p_{vs}(T)$ is the saturation vapor pressure, approximated by the Hyland-Wexler relationship [**ashrae2001**]:

$$p_{vs}(T) = \exp\left[\frac{C_1}{T} + C_2 + C_3T + C_4T^2 + C_5T^3 + C_6\ln(T)\right] \tag{2.38}$$

Sorption isotherms

The relationship between capillary pressure and liquid saturation is described by the Van Genuchten model [**vangenuchten1980**, **baroghel1999**]:

$$p_c = a \left(S_l^{-b} - 1\right)^{1-1/b} \tag{2.39}$$

where a and b are material parameters determined experimentally. Representative values at ambient temperature are given in Table 2.1.

Table 2.1: Van Genuchten parameters at ambient temperature

	NSC	HPC	NSM	HPM
a [MPa]	18.62	46.94	37.55	96.28
b [-]	2.275	2.060	2.168	1.954

At elevated temperatures, the parameter a is modified to account for changes in surface tension and pore structure [**pesavento2000**]:

$$S_l = \left(\left(\frac{E(T)}{a(T)}\, p_c\right)^{b/(b-1)} + 1\right)^{-1/b} \tag{2.40}$$

where $E(T)$ accounts for the surface tension evolution and $a(T)$ accounts for microstructure changes above 100 °C.

Intrinsic permeability

Permeability is a key parameter governing moisture transport. For a thermo-hydral problem, the following relationship is adopted [**gawin1999**]:

$$K = K_0 \cdot 10^{A_T(T-T_0)} \left(\frac{p_g}{p_{atm}}\right)^{A_p} \tag{2.41}$$

where K_0 is the intrinsic permeability at reference conditions, $A_T = 0.005$ and $A_p = 0.368$.

When mechanical damage is considered, the permeability increases significantly [**gawin2002**]:

$$K = K_0 \cdot 10^{A_T(T-T_0)} \left(\frac{p_g}{p_{atm}}\right)^{A_p} \cdot 10^{A_D \mathcal{D}} \tag{2.42}$$

where $\mathcal{D}$ is the damage parameter and A_D is a material constant. This damage-permeability coupling is a key mechanism in the THM model (see Section 2.4).

Relative permeability

The relative permeabilities of liquid and gas are described by the Van Genuchten relationships [**vangenuchten1980**]:

$$k_{rl} = \sqrt{S_l}\left(1 - \left(1 - S_l^{1/A_S}\right)^{A_S}\right)^2 \tag{2.43}$$

$$k_{rg} = \sqrt{1 - S_l}\left(1 - S_l^{1/A_S}\right)^{2A_S} \tag{2.44}$$

where $A_S = 0.6$.

Effective diffusivity

The diffusivity of vapor in air at a given temperature and pressure is [**perre1990**]:

$$D_{va}(T, p_g) = D_{v0}\left(\frac{T}{T_0}\right)^{B_v}\frac{p_{g0}}{p_g} \tag{2.45}$$

where $D_{v0} = 2.58 \times 10^{-5}$ m^2/s and $B_v = 1.667$. The effective diffusivity in the porous medium accounts for saturation and tortuosity:

$$D = (1 - S_l)^{A_v}\,\phi\,\tau\,D_{va}(T, p_g) \tag{2.46}$$

with the tortuosity:

$$\tau(\phi, S_l) = \phi^{1/3}(1 - S_l)^{7/3} \tag{2.47}$$

Fluid viscosities

The dynamic viscosities are temperature-dependent [**ashrae2001**]. The gas viscosity is a concentration-weighted average:

$$\mu_g = \frac{p_v}{p_g}\mu_v + \frac{p_a}{p_g}\mu_a \tag{2.48}$$

where μ_v and μ_a are linear functions of temperature. The liquid water viscosity decreases strongly with temperature [**thomas1995**]:

$$\mu_l = \mu_{l0}\exp\left(\frac{-\alpha_l(T - T_0)}{T}\right) \tag{2.49}$$

with $\mu_{l0} = 1.002 \times 10^{-3}$ Pa·s and $\alpha_l = 0.6612$.

Porosity evolution

The porosity increases with temperature due to microstructural changes [**schneider1989**]:

$$\phi = \phi_0 + A_\phi (T - T_0) \tag{2.50}$$

where ϕ_0 is the initial porosity and A_ϕ is a material constant.

2.2.4 Pore pressure build-up

When concrete is heated, evaporation and dehydration produce vapor that raises the pore pressure. Part of this vapor migrates toward the cooler interior, where it recondenses and forms a quasi-saturated zone — the *moisture clog*. This layer blocks further gas transport, causing significant pressure build-up behind the drying front. In dense concretes such as HPC, the low intrinsic permeability K_0 further limits gas evacuation.

When the pore pressure exceeds the tensile strength, it can trigger explosive spalling (Section 2.5.2).

2.3 Mechanical behavior

[REVIEW]

The mechanical behaviour is modelled by coupling linear elasticity with continuum damage mechanics (Mazars model), described in the following subsections.
The TH and mechanical problems are solved with a staggered (sequential) solution strategy: the TH subsystem is solved first to obtain p_g, p_c, T, which are then passed as known loading to the mechanical step.
The specific coupling method between the TH and mechanical solvers has not yet been finalized and will be detailed and modified in Chapter 2 and Chapter 3 as the study progresses.

2.3.1 Linear elastic model

In the present work, concrete is modeled as a linear elastic material. The effective stress tensor $\tilde{\boldsymbol{\sigma}}$, defined as the stress acting on the undamaged material, is related to the elastic strain tensor $\boldsymbol{\varepsilon}_e$ through the isotropic Hooke's law:

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}_e \tag{2.51}$$

where $\mathbf{C}$ is the constant fourth-order elastic stiffness tensor, defined by the Young's modulus E and Poisson's ratio ν.

The total strain is decomposed into mechanical and free (stress-free) components, as detailed in Section 2.4.2. Since creep is neglected in this work, the elastic strain reduces to:

$$\boldsymbol{\varepsilon}_e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{sh} \tag{2.52}$$

which enters directly into Hooke's law (Equation (2.51)).

Linear finite element formulation

The spatial discretization of the momentum conservation equation (Equation (2.83)) is carried out using the standard Galerkin finite element method [4].

The continuous displacement field $\mathbf{u}$ is approximated by nodal displacements $\{\mathbf{U}\}$ using shape functions $\mathbf{N}$, and the strain is obtained via the compatibility matrix $\mathbf{B}$ such that $\boldsymbol{\varepsilon} = \mathbf{B}\{\mathbf{U}\}$.

Applying the principle of virtual work, the global system takes the form:

$$[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\} \tag{2.53}$$

where $[\mathbf{K}_M] = \int_V \mathbf{B}^T \mathbb{C}\, \mathbf{B}\, dV$ is the elastic stiffness matrix. Its generalisation to account for damage is given in Equation (2.63).

The load vector $\{\mathbf{F}\}$ receives contributions from external forces and from the thermo-hydric fields: (Section 2.4.1):

$$\{\mathbf{F}\} = \{\mathbf{f}_{ext}\} + \underbrace{\int_V \mathbf{B}^T \mathbb{C}\, \boldsymbol{\varepsilon}^{th}\, dV + \int_V \mathbf{B}^T \mathbb{C}\, \boldsymbol{\varepsilon}^{sh}\, dV}_{\text{strain decomposition (Equation (2.71))}} + \underbrace{\int_V \mathbf{B}^T\, \alpha\, p_s\, \mathbf{I}\, dV}_{\text{effective stress (Equation (2.77))}} \tag{2.54}$$

where $\{\mathbf{f}_{ext}\}$ groups the classical external forces. All thermo-hydric terms are computed from the TH solution at each time step and enter as prescribed loading, detailed in Section 2.4.1.

In Cast3M, this system is assembled using the `RIGI` and `RESO` operators for the linear part. When damage is included, the non-linear iterative solution is handled by `PASAPAS` (Equation (2.64)).

2.3.2 Mazars damage model

Concrete cracks when tensile strains become too large. In this work, this behaviour is represented with the Mazars damage model [**mazars1986**, 9].

Damage parameter and equivalent strain

The model introduces a single scalar damage variable $\mathcal{D} \in [0, 1]$ [**mazars1986**]. The effective stress in the damaged material is:

$$\boldsymbol{\sigma}' = (1 - \mathcal{D})\, \mathbb{C}\, \boldsymbol{\varepsilon}_{el} \tag{2.55}$$

When $\mathcal{D} = 0$, this reduces to Hooke's law (Equation (2.51)). The damage variable is driven by the equivalent strain, computed from the positive principal elastic strains:

[REVIEW]

Because concrete cracking is primarily governed by tensile extensions, Mazars introduced an *equivalent strain* (ε_{eq}) to translate any 3D multiaxial strain state into a scalar uniaxial indicator. In the case of a thermo-mechanical loading, it is important to note that only the elastic strain tensor $\boldsymbol{\varepsilon}_e$ contributes to the evolution of damage. The equivalent strain is calculated solely from the positive principal elastic strains ε_i:

$$\varepsilon_{eq} = \sqrt{\langle \boldsymbol{\varepsilon}_e \rangle_+ : \langle \boldsymbol{\varepsilon}_e \rangle_+} = \sqrt{\sum_{i=1}^{3} \langle \varepsilon_i \rangle_+^2} \tag{2.56}$$

where the Macaulay brackets $\langle x \rangle_+$ denote the positive part of x (i.e., $\langle x \rangle_+ = x$ if $x \geq 0$, and 0 if $x < 0$).

Damage initiates when the equivalent strain exceeds the initial damage threshold, denoted as ε_{d0}.

$$\varepsilon_{d0} = \frac{f_t}{E}. \tag{2.57}$$

At the damage onset ($\varepsilon_{eq} = \varepsilon_{d0}$), the damage variable is zero:

$$\mathcal{D} = 0 \qquad \text{for} \qquad \varepsilon_{eq} \leq \varepsilon_{d0}. \tag{2.58}$$

For uniaxial tension, damage onset corresponds to:

$$\sigma_t = f_t. \tag{2.59}$$

In this simplified interpretation, f_t denotes the uniaxial tensile strength of the concrete and E is Young's modulus.

Damage evolution laws for tension and compression

To accurately represent concrete behavior, the total damage D is formulated as a combination of two independent damage variables: D_t for tension and D_c for compression.

When the damage threshold is exceeded ($\varepsilon_{eq} > \varepsilon_{d0}$), the evolution of these two modes is explicitly governed by exponential laws:

$$\mathcal{D}_t = 1 - \frac{(1-A_t)\varepsilon_{d0}}{\varepsilon_{eq}} - A_t \exp\left[-B_t(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{2.60}$$

$$\mathcal{D}_c = 1 - \frac{(1-A_c)\varepsilon_{d0}}{\varepsilon_{eq}} - A_c \exp\left[-B_c(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{2.61}$$

where A_t, B_t, A_c, and B_c are material parameters identified from uniaxial tensile and compressive tests. These parameters modulate the shape of the post-peak softening curves: parameters A define the residual asymptotic stress level, while parameters B control the slope of the stress drop.

The global macroscopic damage $\mathcal{D}$ is then calculated using a linear combination with weighting coefficients α_t and α_c:

$$\mathcal{D} = \alpha_t^{\beta}\mathcal{D}_t + \alpha_c^{\beta}\mathcal{D}_c \tag{2.62}$$

The coefficients α_t and α_c depend on the principal stress state, evaluating the respective contributions of tensile and compressive stresses. In pure tension, $\alpha_t = 1$ and $\alpha_c = 0$; conversely, in pure compression, $\alpha_c = 1$ and $\alpha_t = 0$. The parameter β (typically fixed at 1.06) is a shear-correction coefficient introduced to improve the structural response under shear loadings.

Non-linear equilibrium and iterative solution

In the general case with damage, the stiffness matrix reads:

$$[\mathbf{K}_M] = \int_V \mathbf{B}^T (1-\mathcal{D})\,\mathbb{C}\,\mathbf{B}\,dV \tag{2.63}$$

When $\mathcal{D} = 0$ (intact material), this reduces to the standard linear elastic stiffness matrix $\int_V \mathbf{B}^T\mathbb{C}\,\mathbf{B}\,dV$.

Due to damage evolution, the constitutive relation becomes non-linear and Equation (2.53) cannot be solved in a single step. The equilibrium is enforced iteratively using a Newton-Raphson-type algorithm [4]. At each iteration i, the stiffness matrix (Equation (2.63)) is evaluated with the current damage $\mathcal{D}^i$, and the correction is obtained from:

$$[\mathbf{K}_M]^i\{\delta\mathbf{U}\}^{i+1} = \{\mathbf{F}\} - \{\mathbf{R}\}^i \tag{2.64}$$

where:

- $\{\mathbf{R}\}^i = \int_V \mathbf{B}^T\boldsymbol{\sigma}^i\, dV$ is the internal force vector, with $\boldsymbol{\sigma}^i = (1-\mathcal{D}^i)\,\mathbb{C}\,\boldsymbol{\varepsilon}^i_{el}$;
- $\{\mathbf{F}\}$ is the external load vector including all thermo-hydric contributions (Equation (2.54)).

The displacements and damage are updated until convergence: $\|\{\mathbf{F}\} - \{\mathbf{R}\}^i\| < \epsilon\, F_{ref}$. In Cast3M, this iterative procedure is handled by the `PASAPAS` operator.

2.3.3 Mesh dependency and fracture energy regularization

A well-known drawback of local softening models such as Mazars' is the pathological mesh dependency: softening causes strain localization in a band whose width depends on the element size, leading to vanishing energy dissipation upon mesh refinement [**bazant1976**].

Two main families of regularization exist in the literature:

- **Nonlocal averaging:** the local equivalent strain ε_{eq} is replaced by a weighted spatial average $\bar{\varepsilon}_{eq}$ over a characteristic volume controlled by an internal length l_c [**pijaudier1987**, **giry2011**]. This is the approach adopted by [5].
- **Fracture energy regularization (Hillerborg):** the softening law parameters are adjusted element-by-element so that each element dissipates the same fracture energy G_f per unit crack area, regardless of its size [**hillerborg1976**]. This is the approach adopted in the present work, as it is simpler to implement in Cast3M and does not require additional nonlocal operators.

Principle of fracture energy regularization

Physically, the fracture energy G_f [**hillerborg1976**] represents the energy per unit area required to propagate a crack. In the continuum damage framework, cracking is smeared over the element volume. A characteristic length L_e bridges the two descriptions: in Cast3M (using `@LONGEF`), $L_e = V^{1/D}$ where D is the spatial dimension.

The total dissipated energy per unit crack area is:

$$G_f = G_{elastic} + G_{softening} \tag{2.65}$$

Elastic energy ($G_{elastic}$)

The elastic energy density up to damage onset is:

$$g_{elastic} = \frac{1}{2} f_t \, \varepsilon_{d0} = \frac{f_t^2}{2E_0} \tag{2.66}$$

so that $G_{elastic} = g_{elastic} \, L_e = \frac{f_t^2}{2E_0} \, L_e$, and the energy available for softening is:

$$G_{softening} = G_f - \frac{f_t^2}{2E_0} \, L_e \tag{2.67}$$

Regularization of B_t

Assuming $A_t \approx 1$ (brittle tension), the post-peak stress simplifies to $\sigma \approx f_t \exp[-B_t(\varepsilon - \varepsilon_{d0})]$, whose integral gives $g_{softening} = f_t/B_t$. Equating $G_{softening} = g_{softening} \cdot L_e$ yields the modified parameter:

$$B_t^{mod} = \frac{f_t \, L_e}{G_f - \frac{f_t^2}{2E_0} \, L_e} \tag{2.68}$$

This ensures that every element dissipates exactly G_f regardless of mesh size, making the global response objective with respect to spatial discretization.

2.4 Coupling mechanisms

The thermal, hydric, and mechanical behaviors presented in the previous sections are not independent: they interact through a network of coupling mechanisms. This section identifies and formulates the two main coupling paths that govern the response of concrete exposed to fire.

2.4.1 Thermo-Hydric (TH) coupling mechanisms

The thermal and hydric fields are coupled in both directions: temperature drives phase changes and modifies transport properties, while fluid flow and phase changes affect the energy balance.

Temperature as a driver of hydric phenomena

Temperature enters the hydric conservation equations (Equations (2.33) and (2.34)) through two pathways:

- **Mass source terms:** evaporation and dehydration (Section 2.2.2) are thermally activated processes that inject vapor into the pore network, appearing as source terms $\dot{m}_{vap}$ and $\dot{m}_{dehyd}$ in the mass balance equations;
- **Transport coefficients:** most constitutive parameters are temperature-dependent — permeability $K(T)$, porosity $\phi(T)$, fluid viscosities $\mu(T)$, saturating vapor pressure $p_{vs}(T)$, and diffusivity $D(T)$ — so the coefficients of the mass conservation equations evolve as the thermal field changes.

Energy conservation equation

The energy conservation equation governs the temperature field within the porous medium. Similar to the heat equation in Section 2.1.2, it includes a storage term, a conductive flux, and additionally accounts for advective heat transport and phase-change energy terms [5]:

$$\rho C_p \frac{\partial T}{\partial t} + \underbrace{(m_l C_l \mathbf{v}_{l-s} + m_g C_g \mathbf{v}_{g-s}) \cdot \nabla T}_{\text{advection}} + \nabla \cdot (-\lambda \cdot \nabla T) = \underbrace{-H_{vap}\, \dot{m}_{vap} + H_{dehyd}\, \dot{m}_{dehyd}}_{\text{phase-change energy}} \tag{2.69}$$

Compared to the classical heat equation Equation (2.5) $\rho C_p \frac{\partial T}{\partial t} - \nabla \cdot (\lambda \nabla T) = 0$, two additional contributions appear:

- **Advective heat transport:** The bulk flow of liquid water and gas, driven by pressure gradients, transport enthalpy through the material. This term redistributes energy along the flow paths and can be significant when permeability and pressure gradients are large.
- **Phase-change energy (right-hand side):** Concrete does not self-generate heat; these terms represent energy consumed by evaporation ($-H_{vap}\, \dot{m}_{vap}$, heat sink since $\dot{m}_{vap} > 0$) and dehydration ($+H_{dehyd}\, \dot{m}_{dehyd}$, also a heat sink since $\dot{m}_{dehyd} < 0$). Both locally slow the temperature rise.

After replacing the mass fluxes by Darcy's law (Equations (2.21) and (2.22)), the energy conservation takes its final coupled form:

$$\begin{aligned} \rho C_p \frac{\partial T}{\partial t} - K \left(C_l \frac{\rho_l k_{rl}}{\mu_l} (\nabla p_g - \nabla p_c) + C_g \frac{\rho_g k_{rg}}{\mu_g} \nabla p_g \right) \cdot \nabla T \\ -\nabla \cdot (\lambda \cdot \nabla T) = -H_{vap}\, \dot{m}_{vap} + H_{dehyd}\, \dot{m}_{dehyd} \end{aligned} \tag{2.70}$$

where the overall heat capacity ρC_p (Equation (2.7)) and thermal conductivity λ (Equation (2.10)) are defined in Section 2.1.2.

Thus, the thermal and hydric fields are *strongly coupled*: temperature modifies the source terms and transport coefficients in the moisture equation, while moisture content and fluid flow modify the heat capacity, conductivity, and introduce latent heat and advection terms in the energy equation.

2.4.2 Thermo-hydric input to the mechanical field

The mechanical behaviour of heated concrete is driven by the thermal and hydric fields. The coupling is *one-way*: the converged TH fields (T, p_g, p_c) serve as prescribed loading to the mechanical problem at each time step. The feedback from mechanics to TH through damage-enhanced permeability is discussed in Section 2.5.1.

The TH fields enter the mechanical problem through two mechanisms: free strains and pore pressure loading, detailed in the following subsections.

Strain decomposition

When a mechanically loaded concrete element is exposed to high temperature, the total strain is decomposed as a superposition of components of different origins [5]:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_{\sigma} + \boldsymbol{\varepsilon}_{th} + \boldsymbol{\varepsilon}_{sh} \tag{2.71}$$

where:

- $\boldsymbol{\varepsilon}_{\sigma}$ represents the *mechanical strains* induced by external loading (elastic strain, creep strain, etc.);
- $\boldsymbol{\varepsilon}_{th}$ represents the *thermal dilation strain*, driven by the temperature field;
- $\boldsymbol{\varepsilon}_{sh}$ represents the *capillary shrinkage strain*, driven by the capillary pressure field.

In general, the mechanical strain includes elastic and creep contributions:

$$\boldsymbol{\varepsilon}_{\sigma} = \boldsymbol{\varepsilon}_{el} + \boldsymbol{\varepsilon}_{creep} \tag{2.72}$$

Since creep is neglected in this work, $\boldsymbol{\varepsilon}_{creep} = \mathbf{0}$ and thus $\boldsymbol{\varepsilon}_{\sigma} = \boldsymbol{\varepsilon}_{el}$

Thermal dilation

The thermal dilation strain is isotropic and produces a uniform expansion of the material, which is expressed in incremental form as [5]:

$$d\boldsymbol{\varepsilon}_{th} = \beta_s(T)\, dT\, \mathbf{I} \tag{2.73}$$

where $\mathbf{I}$ is the second-order identity tensor and $\beta_s(T)$ is the thermal dilation coefficient, which for concrete can be approximated as a linear function of temperature:

$$\beta_s(T) = A_{B1}\,(T - T_0) + A_{B0} \tag{2.74}$$

with A_{B0} and A_{B1} being material constants.

Capillary shrinkage

The drying process modifies the pore pressure state, which in turn induces a volumetric deformation of the solid skeleton. The capillary shrinkage strain is expressed as [5]:

$$d\boldsymbol{\varepsilon}_{sh} = \frac{\alpha}{K_T}\,\frac{dp_s}{dt}\,\mathbf{I} \tag{2.75}$$

where α is the Biot coefficient, K_T is the bulk modulus of the drained skeleton, and p_s is the average pore pressure acting on the solid skeleton, defined as:

$$p_s = p_g - S_l\, p_c = S_g\, p_g + S_l\, p_l \tag{2.76}$$

This expression represents the saturation-weighted average of the gas and liquid pressures. Since p_g and p_c are direct outputs of the TH problem, p_s is entirely determined by the TH solution.

Effective stress and pore pressure loading

For a porous medium saturated by multiple fluids, the total stress is decomposed into the part carried by the solid skeleton and the part carried by the pore fluids. Following [**lewis1998**, 5], the effective stress is defined as:

$$\boldsymbol{\sigma}' = \boldsymbol{\sigma}^{Total} + \alpha\, p_s\, \mathbf{I} \tag{2.77}$$

where $\boldsymbol{\sigma}^{Total}$ is the total stress tensor, α is Biot's coefficient, and p_s is the average pore pressure (Equation (2.76)).

The effective stress $\boldsymbol{\sigma}'$ is the stress that governs the mechanical behaviour of the skeleton. In the finite element implementation, this relation is not applied directly; instead, the pore pressure term $\alpha\, p_s$ enters the load vector as an external loading (Equation (2.54)), and the effective stress is recovered from the elastic strain after solving for displacements:

$$\boldsymbol{\sigma}' = (1 - \mathcal{D})\,\mathbb{C}\,\boldsymbol{\varepsilon}_{el} \tag{2.78}$$

where $\boldsymbol{\varepsilon}_{el} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{sh}$ (Equation (2.52)). This effective stress is the input to the Mazars damage criterion (**??**). In heated concrete, the build-up of gas pressure (Section 2.2.4) increases p_s in the load vector, which shifts $\boldsymbol{\sigma}'$ toward tension and promotes damage when f_t is exceeded.

2.5 THM model and spalling

This section assembles the governing equations of the thermo-hydro-mechanical (THM) model from the conservation laws and constitutive relations developed in the previous sections. The coupled system is then used to explain how spalling arises from the interaction between thermal, hydric, and mechanical fields.

2.5.1 Full THM governing formulation

The state of concrete at high temperature is described by four primary variables [5]:

$$\mathcal{U} = \{p_g,\ p_c,\ T,\ \mathbf{u}\} \tag{2.79}$$

where p_g is the gas pressure, p_c is the capillary pressure, T is the temperature, and $\mathbf{u}$ is the displacement vector.

The model consists of four conservation equations, each associated with one primary variable. Their final forms, after substitution of Darcy's law, Fick's law, and the constitutive relations, are as follows.

Dry air conservation

$$\begin{aligned}
&\phi(1-S_l)\left(\frac{\partial \rho_a}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial \rho_a}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial \rho_a}{\partial p_g}\frac{\partial p_g}{\partial t}\right) + (1-S_l)\rho_a\frac{\partial \phi}{\partial T}\frac{\partial T}{\partial t} \\
&- \phi\rho_a\left(\frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t}\right) - \nabla\cdot\left(\frac{K\rho_a k_{rg}}{\mu_g}\nabla p_g\right) - \nabla\cdot\left(D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_a}{p_g}\right) = 0
\end{aligned} \tag{2.80}$$

Water species conservation

$$
\begin{aligned}
&(S_l\rho_l - (1-S_l)\rho_v)\frac{\partial \phi}{\partial T}\frac{\partial T}{\partial t} + S_l\phi\frac{\partial \rho_l}{\partial T}\frac{\partial T}{\partial t} + \phi(\rho_l - \rho_v)\left(\frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t}\right)\\
&+ (1-S_l)\phi\left(\frac{\partial \rho_v}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial \rho_v}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial \rho_v}{\partial p_g}\frac{\partial p_g}{\partial t}\right)\\
&- \nabla\cdot\left(\frac{K\rho_l k_{rl}}{\mu_l}(\nabla p_g - \nabla p_c)\right) - \nabla\cdot\left(\frac{K\rho_v k_{rg}}{\mu_g}\nabla p_g\right) - \nabla\cdot\left(D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_v}{p_g}\right) = -\dot{m}_{dehyd}
\end{aligned}
\tag{2.81}
$$

Energy conservation

$$
\begin{aligned}
\rho C_p\frac{\partial T}{\partial t} - K\left(C_l\frac{\rho_l k_{rl}}{\mu_l}(\nabla p_g - \nabla p_c) + C_g\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g\right)\cdot\nabla T - \nabla\cdot(\lambda\cdot\nabla T)\\
= -H_{vap}\,\dot{m}_{vap} + H_{dehyd}\,\dot{m}_{dehyd}
\end{aligned}
\tag{2.82}
$$

Momentum conservation

By neglecting inertial forces:

$$\nabla\cdot\boldsymbol{\sigma}^{Total} + \rho\,\mathbf{g} = \mathbf{0} \tag{2.83}$$

where $\mathbf{g}$ is the gravitational acceleration and the total density is:

$$\rho = (1-\phi)\rho_s + \phi\,(S_l\rho_l + (1-S_l)\rho_g) \tag{2.84}$$

The total stress $\boldsymbol{\sigma}^{Total}$ is related to the effective stress via Bishop's relation (Equation (2.77)), which in turn depends on the primary variables p_g and p_c.

The discretisation of this equation into the global system $[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\}$ is presented in Section 2.3.

Remark: The governing equations presented in this chapter — dry air conservation, water species conservation, energy conservation, and momentum conservation — together with the constitutive laws, form a complete THM model for concrete exposed to fire. These equations are continuous partial differential equations defined on the domain Ω. Their numerical solution requires spatial and temporal discretization, which is the subject of the following Chapter 3.

2.5.2 Spalling mechanisms

Spalling is defined as the violent or non-violent detachment of layers or pieces of concrete from the surface of a structural element during or after exposure to rapidly rising temperatures. It may lead to premature structural failure through reinforcement exposure and cross-section reduction. Several catastrophic tunnel fires (Channel Tunnel 1996, Mont-Blanc 1999, St. Gotthard 2001) have illustrated the severity of this phenomenon [5].

Despite extensive experimental research, the mechanisms behind spalling are not yet fully understood and no general consensus on a single theory has been reached. The two principal mechanisms most broadly accepted are presented below, followed by their combined action.

Thermal gradient mechanism

[REVIEW] This section is currently under development and will be refined as the study progresses.

When one face of a concrete member is exposed to fire, a steep temperature gradient develops through the thickness. The resulting differential thermal dilation produces (see Figure **??**):

1. **Compressive stresses in the heated cover:** The surface layers tend to expand but are restrained by the cooler interior, generating significant compressive stresses parallel to the heated surface.
2. **Tensile stresses in the interior:** The restraint of the expanding surface layers induces tensile stresses in deeper layers of the material.
3. **Fracture and energy release:** When the tensile stress exceeds the tensile strength f_t, cracks develop parallel to the surface. The sudden release of elastic energy stored in the compressed surface layers can drive the explosive ejection of concrete fragments.

High-performance concrete (HPC) is particularly prone to this mechanism due to its higher brittleness: more elastic energy is stored before failure, leading to a more violent release.

Pore pressure mechanism (moisture clog)

[REVIEW] This section is currently under development and will be refined as the study progresses.

The sequence of events is (see Figure **??**):

1. **Formation of a dry zone:** As the surface is heated, free water evaporates and chemically bound water is released as vapor through dehydration. A dry zone forms near the heated face.

2. **Vapor migration:** Due to pressure gradients, part of the vapor is evacuated toward the heated face, while part is driven toward the cooler interior.

3. **Condensation and moisture clog:** The inward-migrating vapor condenses when it reaches regions where the thermodynamic conditions are satisfied ($T < T_{sat}$). This process continues until a quasi-saturated zone is formed, known as the *moisture clog*, which acts as an impermeable barrier preventing further vapor migration toward the cold side.

4. **Pressure build-up:** Behind the moisture clog, the gas pressure p_g increases sharply as continued evaporation feeds vapor into a region from which it cannot escape. In addition, the thermal dilation of liquid water in saturated pores also contributes to pressure build-up.

5. **Tensile failure:** When the pore pressure exceeds the tensile strength of the material, cracks develop parallel to the surface, potentially leading to explosive spalling.

The key material parameters governing this mechanism are the *intrinsic permeability* K_0 and the *initial moisture content*. Low permeability hinders vapor evacuation, while high moisture content provides more water to be evaporated, both promoting higher pore pressures.

Combined mechanism

[REVIEW] This section is currently under development and will be refined as the study progresses.

In practice, explosive spalling generally occurs under the *combined action* of pore pressures and thermal stresses. The spalling condition can be expressed schematically as:

$$\sigma_{thermal}(T) + \sigma_{pore}(p_g, p_c, S_l) \geq f_t(T) \tag{2.85}$$

where $\sigma_{thermal}$ represents the thermal stress from differential dilation, σ_{pore} represents the stress from pore pressure loading (via effective stress), and $f_t(T)$ is the temperature-degraded tensile strength. When the sum of stresses exceeds the tensile strength, cracks develop parallel to the heated surface, often leading to violent failure of the fire-exposed region.

This combined criterion highlights why the complete THM model is needed: neither a purely thermal analysis (T only) nor a purely hydral analysis (p_g, p_c only) can capture the full picture.

Chapter 3

Numerical implementation

[REVIEW] This chapter is currently empty and is intended to outline the proposed structure of the thesis.

This chapter describes the numerical solution of the THM governing equations developed in Chapter 2. The implementation proceeds incrementally: the thermal, hydric, and mechanical models are first solved separately, then coupled to form the full THM system. All implementations are carried out in the finite element code Cast3M.

3.1 Discretization

3.1.1 Spatial discretization

[REVIEW]

The specific coupling method mentioned in Section 2.3 between the TH and mechanical solvers has not yet been finalized and will be updated in Chapter 2 and Chapter 3 as the study progresses.

Applying the standard Galerkin finite element method to the three TH conservation equations (Equations (2.80) to (2.82)) yields the semi-discrete system [5]:

$$\mathbf{C}\,\dot{\mathbf{Y}} + \mathbf{K}\,\mathbf{Y} = \mathbf{f} \tag{3.1}$$

where $\mathbf{Y} = \{\bar{p}_g,\, \bar{p}_c,\, \bar{T}\}$ is the vector of nodal TH unknowns. The operators $\mathbf{C}$ (capacity),

$\mathbf{K}$(conductivity), and $\mathbf{f}$ (forcing) are 3×3 block matrices:

$$\mathbf{K} = \begin{pmatrix} K_{gg} & K_{gc} & K_{gt} \\ K_{cg} & K_{cc} & K_{ct} \\ K_{tg} & K_{tc} & K_{tt} \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} C_{gg} & C_{gc} & C_{gt} \\ 0 & C_{cc} & C_{ct} \\ 0 & C_{tc} & C_{tt} \end{pmatrix}, \quad \mathbf{f} = \begin{pmatrix} f_g \\ f_c \\ f_t \end{pmatrix} \tag{3.2}$$

The off-diagonal entries K_{ij}, C_{ij} $(i \neq j)$ encode the coupling between the thermal and hydric fields. Detailed expressions of each block are given in [5], Appendix A.

The mechanical problem (Equation (2.83)) is solved in a staggered manner: at each time step, the converged TH fields $\bar{T}$, $\bar{p}_g$, $\bar{p}_c$ enter as prescribed loading. The discrete equilibrium (Equation (2.53)) reads:

$$[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\} \tag{3.3}$$

where $\mathbf{K}_M$ (Equation (2.63)) contains the damaged stiffness $(1-\mathcal{D})\,\mathbb{C}$ — the displacement-dependent part that remains on the left-hand side — and $\mathbf{F}$ (Equation (2.54)) collects the TH-prescribed loads ($\boldsymbol{\varepsilon}^{th}$, $\boldsymbol{\varepsilon}^{sh}$, $\alpha\, p_s$) that do not depend on $\mathbf{U}$.

3.1.2 Temporal discretization

[REVIEW] This section is currently under development and will be refined as the study progresses.

The time integration of Equation (3.1) is carried out using the Wilson-θ method [**lewis1998**, **zienkiewicz2013**], which evaluates the equation at an intermediate time $t_{n+\theta} = t_n + \theta\,\Delta t$ with $\theta \in [0, 1]$. This transforms the semi-discrete system into a fully discrete algebraic system at each time step:

$$\tilde{\mathbf{K}}\,\Delta\mathbf{Y}^{n+1} + \mathbf{K}\,\mathbf{Y}^{n} = \mathbf{f}^{n+\theta} \tag{3.4}$$

where:

$$\Delta\mathbf{Y}^{n+1} = \mathbf{Y}^{n+1} - \mathbf{Y}^{n}, \qquad \tilde{\mathbf{K}} = \frac{\mathbf{C}}{\Delta t} + \theta\,\mathbf{K} \tag{3.5}$$

The parameter θ controls the scheme: $\theta = 0$ gives the explicit forward Euler method, $\theta = 1$ the fully implicit backward Euler, and $\theta = 0.5$ the Crank-Nicolson scheme. In practice, $\theta = 1$ is used for unconditional stability.

Since the capacity and conductivity matrices depend nonlinearly on the unknowns ($\mathbf{C} = \mathbf{C}(\mathbf{Y})$, $\mathbf{K} = \mathbf{K}(\mathbf{Y})$), Equation (3.4) is solved iteratively using the Newton-Raphson method within each time step.

3.1.3 Solution strategy

[REVIEW] This section is currently under development and will be refined as the study progresses.

The TH subsystem (Equation (3.4)) is solved *monolithically*: all three unknowns p_g, p_c, T are updated simultaneously within a single Newton-Raphson loop at each time step.

The mechanical problem (Equation (3.3)) is solved in a *staggered* manner: at each time step, the converged TH solution provides $\bar{T}$, $\bar{p}_g$, $\bar{p}_c$, which are used to assemble $\mathbf{F}$ (Equation (2.54)) and solve for $\mathbf{U}$. The damage variable $\mathcal{D}$ computed from the Mazars model feeds back to the intrinsic permeability via Equation (2.42), creating a weak M→H coupling updated at the next time step.

3.2 Model description and parameters

This geometry is chosen as a representative benchmark for validating the THM implementation against the experimental results presented in Chapter 4, obtained from an experimental campaign conducted under the supervision of S. Dal Pont. The model geometry, material properties, and boundary conditions are described in the following subsections.

3.2.1 Geometry

This section is currently under development and will be refined as the study progresses.

Sample mesh code:

```
OPTI ECHO 0;
OPTI 'DIME' 2 'ELEM' 'QUA4' ;

* 1) GEOMETRY AND MESH DEFINITION

*GEOMETRY DEFINITION
LONG    = 3.;
HAUT    = 0.5;

*NUMBER OF MESH SEGMENT
NLONG   = 35;
NHAUT   = 5;

*POINTS CREATION
```

```
PA      = 0. 0.;
PB      = LONG 0.;
PC      = LONG HAUT;
PD      = 0. HAUT;
TRAC (PA ET PB ET PC ET PD);

*MESH LINES CREATION
LAB     = DROI NLONG PA PB;
LBC     = DROI NHAUT PB PC;
LCD     = DROI NLONG PC PD;
LDA     = DROI NHAUT PD PA;

*MESH SURFACE CREATION
MAIL1   = DALL LAB LBC LCD LDA;
TRAC MAIL1;
```

3.2.2 Material properties

3.2.3 Boundary conditions and loadings

This section is currently under development and will be refined as the study progresses.

3.3 Implementation of the thermal model

3.3.1 Stationary linear thermal analysis

This section is currently under development and will be refined as the study progresses.

3.3.2 Transient linear and non-linear thermal analysis

This section is currently under development and will be refined as the study progresses.

3.4 Implementation of the hydric model

3.4.1 Drying

This section is currently under development and will be refined as the study progresses.

3.4.2 Shrinkage

This section is currently under development and will be refined as the study progresses.

3.5 Implementation of the mechanical model

3.5.1 Elastic linear mechanical model

This section is currently under development and will be refined as the study progresses.

3.5.2 Mazars model

This section is currently under development and will be refined as the study progresses.

3.6 Implementation of the TH coupled model

This section is currently under development and will be refined as the study progresses.

3.7 Implementation of the THM coupled model

This section is currently under development and will be refined as the study progresses.

3.8 Numerical results

This section is currently under development and will be refined as the study progresses.

Chapter 4

Experimental data and Model validation

[REVIEW] This chapter is currently empty and is intended to outline the proposed structure of the thesis.

4.1 Experimental data

4.1.1 Experimental tests and material properties

This section is currently under development and will be refined as the study progresses.

4.1.2 Thermal, hydraulic and mechanic loadings

This section is currently under development and will be refined as the study progresses.

4.1.3 Measured data

This section is currently under development and will be refined as the study progresses.

4.2 Validation

4.2.1 Comparison between numerical and experimental results

This section is currently under development and will be refined as the study progresses.

4.2.2 Discussions on model performance

This section is currently under development and will be refined as the study progresses.

Chapter 5

Probabilistic analysis of the THM model

[REVIEW] This chapter is currently a draft version and will undergo further review and revision.

5.1 Sources of uncertainty and concrete heterogeneity

5.1.1 Motivation for probabilistic assessment

In civil engineering, a structure is considered safe when it remains fit for its intended use throughout its service life [2]. Deterministic models have traditionally been used to evaluate structural safety. However, the gap between a real physical system and a simplified numerical model, along with the unforeseeable nature of future loads and environmental conditions, inherently leads to uncertainty [1].

To manage these risks, modern engineering codes such as the Eurocodes adopt a probabilistic format [2]. This allows the safety of a structure to be explicitly quantified through its probability of failure P_f (Section 5.6.1) and its reliability $R = 1 - P_f$. By introducing random variables into mechanical models and propagating them through structural analyses (Section 5.2), objective decisions can be made based on both the probability of an event and the severity of its consequences [1].

5.1.2 Classification of uncertainties

According to reliability theory, uncertainties can be classified into two categories [2, 1]:

- **Aleatoric uncertainties** arise from the natural randomness of the environment and the spatial or temporal variability of material properties. They cannot be reduced; they can only be quantified and accounted for in the design process [1].
- **Epistemic uncertainties** stem from incomplete knowledge or insufficient data. They include *model uncertainties* (discrepancy between mathematical models and actual physical processes) and *statistical uncertainties* (limited or inaccurate data). Unlike aleatoric uncertainties, they can be reduced by improving models and gathering more data [1].

5.1.3 Concrete heterogeneity and material variability

A critical source of aleatoric uncertainty is the inherent heterogeneity of concrete [1, 6]. Its mechanical properties depend on the formation history, casting conditions, internal chemical reactions, and long-term degradation processes [6].

This variability manifests as *continuous* spatial fluctuations (e.g. Young's modulus, permeability, porosity) and *discrete* defects (e.g. micro-cracks, capillary networks) [1]. Consequently, material parameters must be modelled as random variables or random fields rather than deterministic values (Section 5.2).

5.2 Stochastic finite element framework

In structural engineering, material properties such as the Young's modulus, permeability, or thermal conductivity are never perfectly known. As noted by [6], even under strict quality control the compressive strength of high-performance concrete can range from 62 to 100 MPa. Two approaches exist to handle this uncertainty:

1. A *deterministic* analysis assigns a single fixed value to each parameter and produces a unique structural response; it cannot, by itself, quantify the likelihood that a design limit will be exceeded.
2. A *stochastic* (probabilistic) analysis, on the other hand, treats uncertain parameters as random quantities, propagates this uncertainty through the model, and delivers statistical information—mean, variance, probability of failure—about the structural response.

This work adopts the stochastic approach with two levels of description:

1. **Random variable** (Section 5.2.1): each uncertain property is described by a single distribution (mean, standard deviation) applied uniformly over the entire domain. Multiple random variables can be treated simultaneously.

2. **Random field** (Section 5.2.2): the property varies from point to point, subject to a spatial correlation structure characterised by a correlation length L_c.

The Gaussian Quadrature method (Section 5.4) propagates uncertainty at the random variable level, while Monte Carlo simulation (Section 5.3) handles the random field level.

5.2.1 Random variables (PDF, CDF, moments, CoV)

[REVIEW] This section is currently under development and will be refined as the study progresses.

A **random variable** X is a quantity whose value is not fixed but described by a probability law. Instead of saying "$E = 30$ GPa", one writes $E \sim \mathcal{N}(30, 3^2)$ GPa, meaning that the Young's modulus follows a normal (Gaussian) distribution with mean $\mu = 30$ GPa and standard deviation $\sigma = 3$ GPa.

PDF and CDF

The Probability Density Function (**PDF**) $f_X(x)$ describes the relative likelihood of X taking a given value. It is always non-negative and integrates to one over the entire domain.

The Cumulative Distribution Function (**CDF**)$F_X(x) = P(X \leq x)$ gives the probability that X does not exceed a threshold x:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t. \tag{5.1}$$

In reliability analysis, the CDF is used directly to evaluate the probability of exceeding a critical value: $P(X > x_{\mathrm{cr}}) = 1 - F_X(x_{\mathrm{cr}})$ [2].

Two distributions are used in this work:

- **Normal** (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{5.2}$$

- **Lognormal** distribution $\mathrm{LN}(\mu_{\ln}, \sigma_{\ln}^2)$, where $\ln X$ is normally distributed, ensuring $X > 0$:

$$f_X(x) \;=\; \frac{1}{x\,\sigma_{\ln}\sqrt{2\pi}}\,\exp\!\left[-\frac{(\ln x - \mu_{\ln})^2}{2\sigma_{\ln}^2}\right], \qquad x > 0. \tag{5.3}$$

The lognormal distribution is particularly relevant for material properties such as the Young modulus or permeability, which must remain strictly positive [6, 13, 1].

Statistical moments

The distribution of X can be summarised by its **statistical moments** [2], which quantify location, spread, asymmetry, and tail heaviness. In this work, the first two moments are prescribed as *input* to the stochastic analysis, while the higher moments are *computed as output* by the quadrature method (Section 5.4).

Input moments. The **mean** μ_X and **variance** σ_X^2 (or equivalently the standard deviation $\sigma_X = \sqrt{\mathrm{Var}[X]}$) define the centre and spread of the distribution:

$$\mu_X = \int_{-\infty}^{+\infty} x\, f_X(x)\,\mathrm{d}x, \qquad \sigma_X^2 = \int_{-\infty}^{+\infty} (x-\mu_X)^2\, f_X(x)\,\mathrm{d}x. \tag{5.4}$$

Output moments. The **skewness** δ_3 and **kurtosis** δ_4 measure the asymmetry and tail heaviness of the response distribution:

$$\delta_3 = \frac{E[(X-\mu_X)^3]}{\sigma_X^3}, \qquad \delta_4 = \frac{E[(X-\mu_X)^4]}{\sigma_X^4}. \tag{5.5}$$

These four moments play a central role in the Gaussian Quadrature method (Section 5.4), where the response PDF is reconstructed from the computed moments using Winterstein or Pearson approximations [12].

Coefficient of Variation (CoV)

The coefficient of variation quantifies the relative dispersion of X:

$$\mathrm{CoV}_X = \frac{\sigma_X}{\mu_X}. \tag{5.6}$$

It is a dimensionless measure that allows comparison of variability across parameters with different units and magnitudes. For concrete, typical values are $\mathrm{CoV}_E \approx 0.10$–$0.20$ for the Young's modulus [6] and $\mathrm{CoV}_K \approx 0.15$–$0.30$ for permeability [1].

5.2.2 Random fields and correlation structure

[REVIEW] This section is currently under development and will be refined as the study progresses.

Random fields

A random variable assigns a *single* uncertain value to a material property for the entire structure. In reality, concrete properties vary spatially due to heterogeneous microstructure, local variations in the water-to-cement ratio, and non-uniform compaction during casting [6]. A **random field** captures this spatial variability: it assigns an uncertain value to *every point* $\mathbf{x}$ in the domain, not just one value for the whole structure.

Formally, a random field is a collection of random variables indexed by position [13]:

$$H = \big\{H(\mathbf{x},\omega) : \mathbf{x} \in \mathcal{D},\ \omega \in \Omega\big\}, \tag{5.7}$$

where ω denotes one particular realisation drawn from the probability space. For a fixed ω, $H(\cdot,\omega)$ is a deterministic spatial function (one "map" of the property over the structure); for a fixed $\mathbf{x}$, $H(\mathbf{x},\cdot)$ is an ordinary random variable.

A **second-order** random field is fully characterised by two functions [13]:

- the **mean field** $\bar{H}(\mathbf{x}) = E[H(\mathbf{x})]$,
- the **covariance function** $C_H(\mathbf{x},\mathbf{y}) = E\big[\big(H(\mathbf{x}) - \bar{H}(\mathbf{x})\big)\big(H(\mathbf{y}) - \bar{H}(\mathbf{y})\big)\big]$.

The mean field gives the average value at each point; the covariance function measures how strongly the values at two points $\mathbf{x}$ and $\mathbf{y}$ are linked. When both depend only on the separation $\boldsymbol{\tau} = \mathbf{x} - \mathbf{y}$, the field is called **homogeneous** (stationary).

Correlation function and correlation length

When a material property (e.g. Young's modulus) is modelled as a random field, nearby points tend to have similar values while distant points are more different. The **autocorrelation function** $\rho_H(\boldsymbol{\tau})$ quantifies this similarity as a function of distance τ:

$$\rho_H(\boldsymbol{\tau}) = \frac{C_H(\boldsymbol{\tau})}{\sigma_H^2}, \qquad \rho_H(\mathbf{0}) = 1, \tag{5.8}$$

where $\sigma_H^2 = C_H(\mathbf{0})$ is the field variance. $\rho = 1$ means two points have identical values; $\rho \approx 0$ means they are nearly independent.

The rate at which ρ decays is controlled by the **correlation length** $L_c > 0$. Two common models for concrete [13, 8]:

$$\rho_H(\tau) = \exp\left(-\frac{\tau^2}{L_c^2}\right) \text{ (squared-exponential)}, \qquad \rho_H(\tau) = \exp\left(-\frac{|\tau|}{L_c}\right) \text{ (exponential)}. \tag{5.9}$$

Physically, L_c represents the distance beyond which material properties become essentially independent. A small L_c means the material varies rapidly in space (highly heterogeneous); a large L_c means properties change slowly (nearly uniform). This effect is illustrated in **??**.

Remark: In this work, L_c is not treated as a parameter to be varied. A single representative value is adopted from experimental data [6] to account for spatial heterogeneity in the Monte Carlo simulations (**??**). The objective is not to study the effect of L_c itself, but to compare the structural response *with* and *without* spatial variability.

5.2.3 Random field simulation

To use a random field in a finite element model, one must *generate* realisations that respect the prescribed covariance structure, then *assign* the values to elements or integration points. Several families of methods exist [8]:

- **Matrix decomposition (Cholesky).** The covariance matrix $\mathbf{C}$ is factorised as $\mathbf{C} = \mathbf{L}\mathbf{L}^T$ and a correlated realisation is obtained as $\mathbf{h} = \mathbf{L}\mathbf{z}$, where $\mathbf{z}$ is a vector of independent standard normal variates [8]. The method is exact but scales as $\mathcal{O}(n^3)$ in the number of nodes n, making it prohibitive for large meshes.
- **Spectral methods (FFT-based).** Realisations are generated by filtering white noise in the frequency domain using the spectral density corresponding to the target covariance [8]. FFT algorithms are computationally efficient on regular grids, but the method is limited to stationary fields on rectangular domains with periodic boundary conditions.

- **Local Average Subdivision (LAS).** LAS generates the random field by successively subdividing the domain and assigning local averages consistent with the target covariance at each level of refinement [8]. The method is computationally efficient and naturally produces spatially averaged values over elements, making it well-suited for direct use in finite element models. However, it is primarily designed for regular rectangular grids.

- **Series expansion methods.** Methods such as the Karhunen–Loève (KL) expansion parametrise the random field as a finite series of deterministic spatial functions multiplied by independent random variables [13, 6]. They provide a compact representation of the spatial variability but require solving a global eigenvalue problem on the finite element mesh, whose cost increases with mesh size and targeted accuracy.

- **Turning Bands Method (TBM).** The TBM reduces the d-dimensional simulation to a set of one-dimensional stationary processes along random lines (bands) passing through the domain [11, 8]. It requires neither a global matrix factorisation nor an eigenvalue solve, and operates directly on any mesh geometry.

In this work, the TBM method, described in detail in Section 5.3.4, is adopted because it is the method implemented in the Cast3M `ALEA` operator.

5.2.4 Summary: random variable versus random field

Table 5.1 summarises the key differences between the two levels of stochastic description and their implications for the numerical methods used in this work.

Table 5.1: Comparison of random variable and random field stochastic descriptions.

Aspect	**Random variable**	**Random field**
Spatial variation	None (uniform)	Yes (spatially heterogeneous)
Characterisation	PDF: mean μ, std σ	Mean field $\bar{H}(\mathbf{x})$, covariance $C_H(\mathbf{x}, \mathbf{y})$
Key parameter	CoV	Correlation length L_c
Stochastic level	First-order	Second-order
Simulation method	Gaussian Quadrature	TBM + Monte Carlo
Cast3M operator	`QUADRATU`	`ALEA`
Computational cost	Low (3–7 runs per variable)	High (hundreds to thousands of runs)

5.3 Monte Carlo simulation framework

5.3.1 Principle

Monte Carlo simulation (MCS) estimates the effect of uncertain inputs by running the same deterministic model many times, each time with a different random sample [8]. Each trial consists of three steps:

1. Generate a realisation of the random input (random variable or random field) according to its prescribed distribution.
2. Run the deterministic model (here, the THM finite element solver in Cast3M) to obtain the response.
3. Collect the result; repeat until the output statistics are stable.

Because no assumption is made on the shape of the response, MCS can estimate the *entire* output distribution—unlike linearisation methods such as FORM/SORM [8].

5.3.2 Convergence

The probability of failure is estimated as the fraction of realisations where the limit state is violated:

$$\hat{P}_f = \frac{N_f}{n}, \tag{5.10}$$

where the hat denotes that $\hat{P}_f$ is an estimate of the true P_f, which improves as n increases, and N_f is the number of failures out of n total runs. The more runs, the more accurate the estimate—but the error only decreases as $1/\sqrt{n}$. To halve the error, one must quadruple the number of runs.

At 95% confidence, the required number of runs is [8]:

$$n \;\geq\; \frac{4\,p_f(1-p_f)}{e^2}, \tag{5.11}$$

where e is the acceptable absolute error. For a small failure probability ($p_f \approx 10^{-3}$) with 10% relative accuracy, this gives $n \approx 400{,}000$ runs. When a single THM analysis takes several minutes, this cost becomes prohibitive—motivating the Gaussian Quadrature method (Section 5.4) as a faster alternative.

5.3.3 Transformation of Lognormal to Gaussian distribution

In practical finite element implementations using the Cast3M software, material parameters such as Young's modulus and permeability must remain strictly positive. However, the built-in random field operator `ALEA` generates Gaussian (normal) distributions, necessitating a mathematical transformation [13].

A **lognormal random field** $H(\mathbf{x})$ is therefore adopted: by definition, $\ln H(\mathbf{x})$ is a Gaussian random field, which guarantees $H(\mathbf{x}) > 0$ at every point [6]. The moments of $H(\mathbf{x})$ are related to those of the underlying Gaussian field $G(\mathbf{x}) \sim \mathcal{N}(\mu_{\ln}, \sigma_{\ln}^2)$ by [13]:

$$\begin{cases} \mu_X &= \exp\left(\mu_{\ln} + \dfrac{\sigma_{\ln}^2}{2}\right) \\ \sigma_X^2 &= [\exp(\sigma_{\ln}^2) - 1] \exp(2\mu_{\ln} + \sigma_{\ln}^2) \end{cases} \tag{5.12}$$

where μ_X and σ_X are the physical mean and standard deviation of the material property. Inverting Equation (5.12) yields the Gaussian parameters as functions of the physical moments:

$$\sigma_{\ln}^2 = \ln\bigl(1 + CoV_X^2\bigr)\,, \qquad \mu_{\ln} = \ln(\mu_X) - \tfrac{1}{2}\,\sigma_{\ln}^2, \tag{5.13}$$

where $CoV_X = \sigma_X/\mu_X$ is the coefficient of variation.

The standard computational procedure in Cast3M `ALEA` then consists of two steps:

1. Generate a Gaussian random field $G(\mathbf{x}) \sim \mathcal{N}(\mu_{\ln}, \sigma_{\ln}^2)$ using the Turning Band Method (see Section 5.3.4), with parameters computed from Equation (5.13).

2. Transform pointwise to obtain the lognormal field:

$$H(\mathbf{x}) \;=\; \exp\bigl(G(\mathbf{x})\bigr), \tag{5.14}$$

   which guarantees $H(\mathbf{x}) > 0$ and preserves the correlation structure characterised by L_c (see Section 5.2.2).

This pointwise transformation is exact: the autocorrelation structure of the underlying Gaussian field is preserved in the lognormal field, so the spatial variability characterised by the correlation length L_c remains physically meaningful after transformation [6].

5.3.4 Turning Bands Method (ALEA)

Principle The Turning Bands Method (TBM), introduced by [11], simplifies the generation of a 2-D or 3-D random field by reducing it to a set of 1-D problems.

The idea (**??**): draw L lines through the domain in different directions. Along each line, simulate an independent 1-D random process. To obtain the field value at any point $\mathbf{x}$, project it onto each line and average the results:

$$Z(\mathbf{x}) \;=\; \frac{1}{\sqrt{L}} \sum_{i=1}^{L} Z_i(\mathbf{x} \cdot \mathbf{u}_i)\,, \tag{5.15}$$

where $\mathbf{u}_i$ is the direction of the i-th line and Z_i is the 1-D process along it. The 1-D processes are constructed so that the resulting field reproduces the prescribed covariance structure (Section 5.2.2). In practice, $L = 16$ lines are sufficient for isotropic fields in two dimensions [8].

Implementation in Cast3M The TBM is natively implemented in the Cast3M `ALEA` operator, which takes as input the Gaussian parameters $(\mu_{\ln}, \sigma_{\ln})$ and the correlation length L_c, and returns the simulated field directly as nodal values on the mesh. The lognormal transformation and assignment to `MATE` follow the procedure described in Section 5.3.3.

5.3.5 Integration of spatial variability into the THM model

This section is currently under development and will be refined as the study progresses.

Once the lognormal random field $H(\mathbf{x})$ has been constructed via the Turning Band Method (see Section 5.3.4), it is propagated through the nonlinear THM model using Monte Carlo simulation. Given the computational cost of each THM evaluation, the iterative procedure is fully automated in Cast3M using the `REPETER` operator.

Simulation loop and dynamic material updating

For a predefined number of simulations N_{simu}, the global algorithm performs the following sequential steps within each loop:

1. **Random Field Generation:** A new statistically independent realization of the random field is generated using the `ALEA` operator. For instance, the spatial distribu-

tion of the Young's modulus $\mathbf{E}(x)$ is recalculated based on the assigned log-normal parameters (mean, standard deviation, and correlation length).

2. **Material Updating:** The macroscopic properties of the finite element model are updated. The newly generated random field is projected onto the mesh to form the updated material stiffness matrix using the `MATE` operator.

3. **Non-linear Resolution:** The `PASAPAS` procedure is invoked to solve the multi-physics THM problem from $t = 0$ to the final time t_{fin}, utilizing the updated heterogeneous material properties.

Data extraction and post-processing

This section is currently under development and will be refined as the study progresses

5.4 Gaussian Quadrature framework

Unlike Monte Carlo, which runs thousands of simulations, Gaussian Quadrature evaluates the model at a small number of strategically chosen points and reconstructs the output statistics from a weighted sum [**baldeweck1999**, 12]. The method has three steps, corresponding to three Cast3M operators:

1. `QUADRATU`: compute the evaluation points and weights for each random variable.
2. `PARASTAT`: run the model at each point combination and compute the first four statistical moments (mean, standard deviation, skewness, kurtosis).
3. `PROBDENS`: reconstruct the PDF from these moments and estimate P_f and β.

5.4.1 Step 1: Quadrature points and weights (`QUADRATU`)

Computing any statistical moment of the output $G = g(\mathbf{X})$ requires evaluating an integral:

$$I = \int_{\mathcal{D}} g(x)\, W(x)\, \mathrm{d}x, \tag{5.16}$$

where $W(x) = f_X(x)$ is the PDF of the input variable. Gaussian quadrature approximates this by a weighted sum over K evaluation points [**baldeweck1999**]:

$$I \approx \sum_{k=1}^{K} w_k\, g(x_k), \tag{5.17}$$

where $\{x_k\}$ are the *abscissas* (evaluation points) and $\{w_k\}$ the corresponding *weights*. The optimal abscissas are the roots of orthogonal polynomials associated with W, and a K-point scheme integrates exactly any polynomial of degree up to $2K-1$ [12].

`QUADRATU` automatically computes $\{x_k, w_k\}$ for any prescribed distribution (Normal, Lognormal, Uniform, etc.). The user provides only the distribution type and its parameters.

5.4.2 Step 2: Statistical moments (`PARASTAT`)

For N independent random variables with K points each, the quadrature is applied to each variable separately and combined as a tensor product [12]:

$$\mu_q^{(0)} \approx \sum_{k_1=1}^{K_1} \cdots \sum_{k_N=1}^{K_N} w_{k_1} \cdots w_{k_N} \left[g(x_{k_1}, \ldots, x_{k_N})\right]^q, \tag{5.18}$$

requiring K^N model evaluations in total. For example, with $N=2$ variables and $K=3$ points each: $3^2 = 9$ runs.

`PARASTAT` collects these evaluations and returns the first four moments: mean μ, standard deviation σ, skewness $\delta_3 = \mu_3/\sigma^3$, and kurtosis $\delta_4 = \mu_4/\sigma^4$.

5.4.3 Step 3: PDF reconstruction and reliability (`PROBDENS`)

Once the four moments are known, `PROBDENS` reconstructs the PDF using *three independent methods simultaneously* [12]:

- **Winterstein approximation:** expresses G as a Hermite polynomial transformation of a standard normal variable. Simplest but least accurate of the three methods.
- **Pearson's method:** selects the best-fit distribution family (normal, beta, gamma, etc.) from the moment values. Most accurate, but does not always provide a result.
- **Exponential of polynomials ($n=4$):** models the PDF as $\tilde{f}_G = c\,\exp(Q)$ where Q is a polynomial fitted to the moments. Always provides accurate results.

Each method independently yields a probability of failure P_f and reliability index β. When the three results are consistent with each other, they can be accepted with confidence [12]:

$$P_f = \tilde{F}_G(0), \qquad \beta = -\Phi^{-1}(P_f), \tag{5.19}$$

where $\tilde{F}_G$ is the reconstructed CDF and Φ^{-1} the inverse standard normal CDF. The method provides accurate results for $\beta \lesssim 5$ [12].

5.4.4 Computational cost

The total number of model evaluations is K^N, which grows exponentially with N. For the present study with $N = 2$ and $K = 3$, this gives only 9 runs—compared to thousands for Monte Carlo. However, for large N (e.g. $3^{10} = 59{,}049$ runs), the cost becomes comparable to Monte Carlo. This *curse of dimensionality* limits the method to problems with a small number of random variables ($N \lesssim 10$–15) [12].

5.5 Sensitivity analysis

The THM model involves numerous material parameters. Following [10], 18 key parameters are selected for sensitivity analysis based on their expected influence on the structural response.

5.5.1 Characterization of input parameters

Characterize the chosen input parameters in terms of domain variations and coefficient of variation.

[REVIEW]
Main ideas as follows:

Chosen Random variables

Correlated vs uncorrelated parameters

Physical correlations

5.5.2 Sensitivity indicators and ranking methods

[REVIEW] This section is currently under development and will be refined as the study progresses.

Variance decomposition

For a model $G = g(X_1, \ldots, X_N)$ with N independent random inputs, Hoeffding's decomposition **[sobol1993]** expresses G as a sum of functions of increasing dimension:

$$G \;=\; g_0 \;+\; \sum_{i=1}^{N} g_i(X_i) \;+\; \sum_{i<j} g_{ij}(X_i, X_j) \;+\; \cdots \;+\; g_{1,\ldots,N}(X_1, \ldots, X_N), \tag{5.20}$$

where $g_0 = \mathbb{E}[G]$ is a constant and each term has zero mean. Taking the variance of both sides yields:

$$\mathrm{Var}(G) \;=\; \sum_{i=1}^{N} V_i \;+\; \sum_{i<j} V_{ij} \;+\; \cdots \;+\; V_{1,\ldots,N}, \tag{5.21}$$

where $V_i = \mathrm{Var}[g_i(X_i)]$ is the first-order variance (effect of X_i alone) and $V_{ij} = \mathrm{Var}[g_{ij}(X_i, X_j)]$ is the second-order variance (interaction between X_i and X_j).

Variance-based sensitivity: Sobol indices

Dividing each term in Equation (5.21) by the total variance defines the *Sobol indices* **[sobol1993]**:

$$S_i \;=\; \frac{V_i}{\mathrm{Var}(G)}, \qquad S_{ij} \;=\; \frac{V_{ij}}{\mathrm{Var}(G)}, \tag{5.22}$$

where:

- S_i is the **first-order Sobol index**: fraction of the output variance due to X_i alone.
- S_{ij} is the **second-order Sobol index**: fraction due to the interaction between X_i and X_j, beyond their individual effects.

The **total-order Sobol index** captures the full effect of X_i, including all interactions:

$$S_{T_i} \;=\; S_i \;+\; \sum_{j\neq i} S_{ij} \;+\; \cdots \tag{5.23}$$

By construction, the first-order indices sum to at most one ($\sum_i S_i \leq 1$), and the total-order indices sum to at least one ($\sum_i S_{T_i} \geq 1$). When the model is additive (no interactions), $S_{ij} = 0$ for all pairs and $S_i = S_{T_i}$.

The first-order index S_i can be expressed as a conditional variance:

$$S_i = \frac{\mathrm{Var}_{X_i}\left[\mathbb{E}_{X_{\sim i}}(G \mid X_i)\right]}{\mathrm{Var}(G)}, \tag{5.24}$$

where $X_{\sim i}$ denotes all variables except X_i. Intuitively, $\mathbb{E}(G \mid X_i)$ averages out all other variables: if this conditional mean varies a lot with X_i, then X_i is influential.

A variable with a large S_i (or S_{T_i}) is a *dominant parameter*: it drives most of the output uncertainty and should be characterised with care. A variable with a small index could be treated as deterministic without significant loss of accuracy.

Practical sensitivity approximation

Computing exact Sobol indices requires a large number of model evaluations. A simpler approach, used by the the expansion of Englishquadrature method (Section 5.4), estimates the contribution of each variable one at a time:

1. Fix all variables at their mean except X_i.
2. Run the model at each quadrature point of X_i to obtain the marginal variance $\hat{\sigma}_i^2$.
3. Repeat for every variable.

The **contribution** of X_i to the total output variability is then:

$$C_i = \frac{\hat{\sigma}_i^2}{\displaystyle\sum_{j=1}^{N} \hat{\sigma}_j^2} \times 100\,\% \approx S_i \quad \text{when } S_{ij} \approx 0. \tag{5.25}$$

A variable with $C_i \approx 0\,\%$ has negligible influence and could be treated as deterministic. This approximation is valid when variables are uncorrelated and interaction effects are small. The detailed results are presented in Section 5.5.3.

5.5.3 Identification of dominant parameters

[REVIEW]

Main ideas as follows:

Ranking from most to least influential

Comparison: Quadrature ranking vs Monte Carlo ranking

Physical interpretation of dominant parameters

5.6 Reliability analysis

[REVIEW]

Main ideas as follows:

5.6.1 Probability of failure and reliability index

Limit state function

In the sensitivity analysis (Section 5.5), $G = g(X_1, \ldots, X_N)$ denoted the model output without any notion of safety or failure. In reliability analysis, the same symbol G is given an additional meaning: its *sign* now defines whether the structure is safe or has failed.

Specifically, G is rewritten as a **performance function** (also called limit-state function or safety margin) [2]. A simple example: if R is the material resistance and S is the applied load, then $G = R - S$. The structure is safe when $G > 0$ and fails when $G \leq 0$:

$$G(\mathbf{X}) > 0 \;\Rightarrow\; \text{safety domain } D_s, \qquad G(\mathbf{X}) \leq 0 \;\Rightarrow\; \text{failure domain } D_f. \tag{5.26}$$

In the simple case of a resistance R and a load effect S, the performance function writes $G(R, S) = R - S$ [2].

Probability of failure

The probability of failure is defined as [2]:

$$P_f \;=\; \mathrm{Prob}\big[G(\mathbf{X}) \leq 0\big] \;=\; \int_{D_f} f_{\mathbf{X}}(\mathbf{x})\,\mathrm{d}\mathbf{x}, \tag{5.27}$$

where $f_{\mathbf{X}}$ is the joint PDF of the random input vector. This integral is generally intractable for complex systems, which motivates the use of reliability indices.

Reliability indices

Given P_f from Equation (5.27), the *generalized reliability index* is defined as [12]:

$$\hat{\beta} = -\Phi^{-1}(\hat{P}_f), \tag{5.28}$$

where Φ is the standard normal CDF. This definition applies to both the quadrature method (where $\hat{P}_f$ is obtained from the reconstructed CDF, Section 5.4.3) and Monte Carlo simulation (where $\hat{P}_f$ is the empirical failure ratio, Section 5.3).

Remark: Classical reliability theory also defines the Cornell index $\beta_C = \mu_G/\sigma_G$ and the Hasofer–Lind index β_{HL} (minimum distance to the failure surface in the standard normal space) [2, 1]. The Cornell index is exact only for linear limit states; the Hasofer–Lind index is used in FORM/SORM methods. Neither is employed in this work; for details, the reader is referred to [2, 1].

5.6.2 PDF/CDF-based assessment (Quadrature)

5.6.3 Sampling-based assessment (Monte Carlo)

5.6.4 Effect of spatial heterogeneity on structural safety

The quadrature method models each uncertain parameter as a single random variable, constant over the entire structure (Section 5.4). The Monte Carlo framework, on the other hand, models parameters as random fields discretised via the Turning Band Method, allowing spatial variability within each realisation (Section 5.3).

The key consequence for reliability is that spatial heterogeneity generally leads to *more critical* (conservative) results compared to the homogeneous assumption:

- With random variables (quadrature), the entire structure has the same material property in each realisation — weak zones are compensated by strong zones.
- With random fields (Monte Carlo), local weak zones can form where failure initiates, without being compensated by surrounding material.

This explains why Monte Carlo simulations with spatial variability typically yield higher probabilities of failure (lower β) than the quadrature approach for the same nominal input distributions [6, 8].

Chapter 6

Conclusions and Perspectives

[REVIEW] This chapter is currently empty and is intended to outline the proposed structure of the thesis.

6.1 Main findings

This section is currently under development and will be refined as the study progresses.

6.2 Model limitations and possible improvements

This section is currently under development and will be refined as the study progresses.

6.3 Perspectives for future research

This section is currently under development and will be refined as the study progresses.

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